26/3/26.

T-9. (KNN).

(20) → 5 fold k-fold. CV. $\frac{20}{5} = 4$.

Training (16). validation (4)

∴ The result will be used in validation set.

Customer Id	Age	Actual Income	Shopping frequency	Purchase.
1.	25	50	8	Y
2	45	80	3	N
3	28	55	9	Y
4	50	85	2	N
5	30	60	10	Y
6	48	82	3	N
7.	26	52	8	Y
8.	47	88	2	N
9.	29	58	9	Y
10	46	84	3	N
11	27	54	8	Y
12.	49	86	2	N
	31	62	10	Y
	44	81	3	N
	28	56	9	Y
	51	87	2	N
	26	53	8	Y
	48	83	3	N
	30	59	9	Y
	45	80	3	N

For only 12 values.

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sol:- Fold - 1 (19, 3)

i) Distance for (46, 84, 3)

1. $\sqrt{(x_{test}-x_1)^2+(x_{test}-x_2)^2+(x_{test}-x_3)^2} = 40.24$
2. 4.123 $= \sqrt{(46-45)^2+(84-80)^2+(3-3)^2}$ = N
3. 34.65
4. 4.24 - N
5. 29.68 - Y
6. 2.68 N
7. 30.66
8. 4.24 N
9. 31.6

kNN (k = 5)

2) Distance of (27, 54, 8)

1. $\sqrt{(27-25)^2+(54-50)^2+(8-8)^2} =$ 4.47 Y
2. $\sqrt{(27-45)^2+(54-80)^2+(8-3)^2} =$ 32.01.
3. $\sqrt{(27-28)^2+(54-55)^2+(8-9)^2} =$ 1.732 Y
4. $\sqrt{(27-50)^2+(54-85)^2+(8-2)^2} =$ 39.06.
5. $\sqrt{(27-30)^2+(54-60)^2+(8-10)^2} =$ 7 Y
6. $\sqrt{(27-48)^2+(54-82)^2+(8-3)^2} =$ 35.35
7. $\sqrt{(27-26)^2+(54-52)^2+(8-8)^2} =$ 2.236 Y
8. $\sqrt{(27-47)^2+(54-88)^2+(8-2)^2} =$ 39.89
9. $\sqrt{(27-29)^2+(54-58)^2+(8-9)^2} =$ 4.58 Y

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3) Distance (49, 86, 2)

1. $\sqrt{(49-25)^2+(86-50)^2+(2-8)^2} = 43.68$

2. $\sqrt{(49-47)^2+(86-80)^2+(2-3)^2} = 7.28$ N

3. $\sqrt{(49-28)^2+(86-55)^2+(2-9)^2} = 38.09$

4. $\sqrt{(49-50)^2+(86-85)^2+(2-2)^2} = 1.414$ N

5. $\sqrt{(49-30)^2+(86-60)^2+(2-10)^2} = 33.181$ Y

6. $\sqrt{(49-48)^2+(86-82)^2+(2-3)^2} = 4.24$ N

7. $\sqrt{(49-26)^2+(86-52)^2+(2-8)^2} = 41.48$

8. $\sqrt{(49-47)^2+(86-88)^2+(2-2)^2} = 2.82$ N

9 - $\sqrt{(49-29)^2+(86-58)^2+(2-9)^2} = 35.11$